

JNCD), while values above 3 indicate visible color discrepancies.

We also evaluated color gamut coverage in the sRGB space and overall color volume, which indicates how well a display can reproduce vivid and accurate colors across its entire range.

## Breaking Down the Results

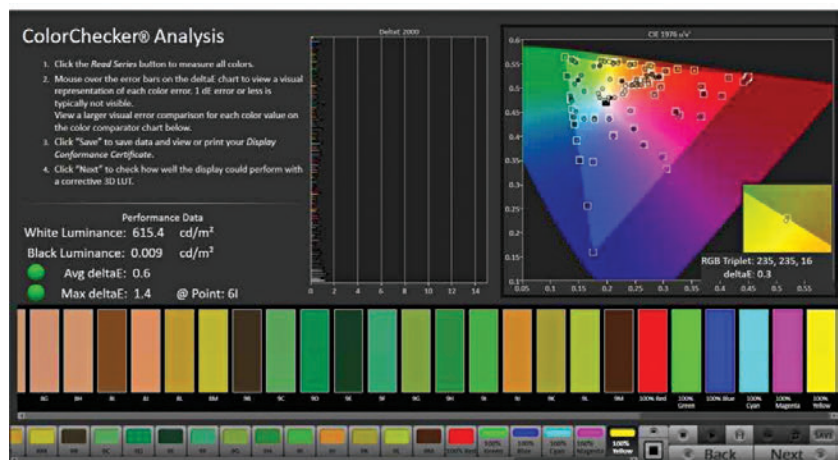
The brightness battle saw the Galaxy S24 Ultra and Pixel 9 Pro XL emerge as clear leaders, both achieving an impressive 2000 nits peak brightness. This significant advantage translates to superior outdoor visibility compared to their competitors.

In terms of color accuracy, the iQOO 13 surprised everyone by leading the pack with an exceptional Delta E value of 0.6. The Realme GT 6 followed closely at 0.9, while the Samsung Galaxy S24 Ultra and Honor Magic6 Pro both achieved a respectable 1.1. The OnePlus 12 showed slightly more deviation with a 1.9 Delta E value.

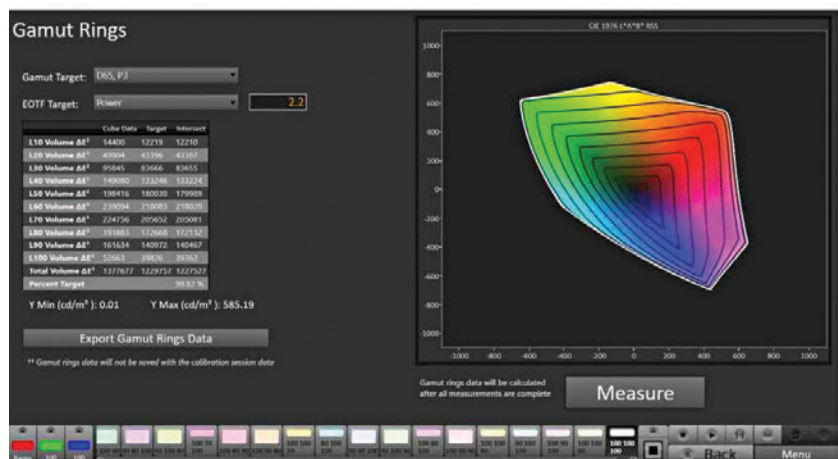
Color gamut coverage testing revealed outstanding performance across the board, with the OnePlus 12 and iQOO 13 achieving perfect 100% sRGB coverage. The remaining devices weren't far behind, demonstrating the high capability of modern smartphone displays.

## Real-World Implications

These technical measurements translate into tangible user experiences. The S24 Ultra and Pixel 9 Pro XL offer the most well-rounded package, excelling



iQOO 13 Colour Checker Analysis



iQOO 13 Colour Gamut Analysis



in both brightness and color accuracy. This means users will enjoy excellent visibility in all lighting conditions while maintaining natural color reproduction.

The iQOO 13's exceptional color accuracy makes it a standout choice for photography enthusiasts and content creators who require precise color representation. Meanwhile, the OnePlus 12's high color gamut coverage and unique gamma tuning of 2.4 in Pro mode makes it particularly suitable for gaming and HDR content consumption in darker environments.

The Honor Magic6 Pro demonstrates consistent performance across all metrics, while the Realme GT6 proves that excellent display quality doesn't necessarily require a premium price tag.

## The Bottom Line

Modern flagship smartphones have largely solved the challenge of color accuracy, but key differentiators remain in areas like brightness, power efficiency, and responsiveness. Premium displays generally offer better HDR performance and more sophisticated tone curve optimization for various lighting conditions.

The key takeaway for consumers is that while all tested devices perform admirably, choosing the right color profile in display settings is crucial for optimal performance. The distinction between good and great now lies in specific use-case requirements rather than overall quality, marking a significant evolution in smartphone display technology. **d**